

Microsoft Fabric Data Engineering Presentation (Simple English)

Slide 1: Introduction

Good Morning Sir,

Today I will explain my understanding of Microsoft Fabric Data Engineering.

In many companies, data comes from different places such as databases, APIs, Excel files, and applications.

Because data is stored in different systems, companies face problems like:

- Data is scattered everywhere.
- Reports take more time to create.
- Managing data becomes difficult.
- Maintenance costs are high.

Microsoft Fabric helps solve these problems by bringing everything into one platform.

It helps us collect, store, process, and analyze data in a single place.

Slide 2: Overall Architecture

The data flow in Microsoft Fabric is simple.

Source Systems → Data Factory → OneLake → Lakehouse → Warehouse → Power BI

First, we collect data from different sources.

Then we store the data in OneLake.

After that, we clean and transform the data in Lakehouse.

Next, we prepare business reports using Warehouse.

Finally, users view reports and dashboards in Power BI.

This creates one central platform for all data needs.

Slide 3: Data Ingestion

Data Ingestion means bringing data into Microsoft Fabric.

Data can come from:

• SQL Server • Oracle • APIs • CSV files • Excel files

We use Data Factory pipelines to move data.

Instead of creating separate pipelines for every table, we create one reusable pipeline.

This saves time and reduces maintenance effort.

For example:

If there are 100 tables, one pipeline can load all 100 tables automatically.

Slide 4: OneLake

OneLake is the main storage area in Microsoft Fabric.

We can think of it as OneDrive for company data.

Benefits:

• All data is stored in one place. • Easy to manage. • Easy to access. • No duplicate copies.

OneLake also supports shortcuts.

This means we can access data from other systems without physically copying it.

This saves storage and reduces costs.

Slide 5: Lakehouse

Lakehouse is where we store and process data.

It combines the benefits of:

• Data Lake • Data Warehouse

Advantages:

- Better performance • Better data management • Easy data processing

Microsoft Fabric stores data in Delta format.

Benefits of Delta format:

- Keeps data accurate. • Prevents data corruption. • Supports version history. • Improves performance.
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Slide 6: Medallion Architecture

Medallion Architecture has three layers.

Bronze Layer

This is the raw data layer.

Data is stored exactly as it comes from the source.

No changes are made.

Silver Layer

This is the cleaned data layer.

Activities include:

- Removing duplicate records • Fixing data types • Handling missing values

Gold Layer

This is the business-ready layer.

Data is prepared for reporting and analytics.

This is the layer used by Power BI reports.

Slide 7: Data Transformation

After loading data, we need to clean and transform it.

We use PySpark notebooks for this work.

Examples:

- Remove duplicates • Convert data types • Create business calculations • Combine multiple datasets

Fabric automatically improves performance using built-in optimization features.

This helps reports run faster.

Slide 8: Fabric Warehouse

Fabric Warehouse is used for SQL reporting.

Users can write SQL queries directly.

Benefits:

- Easy for SQL developers • Fast reporting • Supports large datasets

Business users can access data without using Spark code.

Slide 9: Incremental Loading

Incremental Loading means loading only new data.

Example:

Yesterday we loaded 1 million records.

Today only 10,000 new records are added.

Instead of loading all 1 million records again, we load only the 10,000 new records.

Benefits:

- Faster processing • Lower cost • Better performance
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Slide 10: Security and Governance

Security is very important.

Microsoft Fabric provides:

Workspace Roles

Admin – Full access

Contributor – Can develop pipelines and notebooks

Viewer – Can only view reports

Row Level Security

Users can see only the data they are allowed to see.

Example:

A Hyderabad manager sees only Hyderabad data.

A Chennai manager sees only Chennai data.

This improves data security.

Slide 11: Power BI Direct Lake

Power BI is used for reporting.

Microsoft Fabric provides Direct Lake Mode.

Benefits:

- Real-time reporting • Faster dashboard performance • No need to import data repeatedly

Users can view the latest data instantly.

This is one of the biggest advantages of Microsoft Fabric.

Slide 12: Business Benefits

Microsoft Fabric provides:

- One platform for everything • Faster reporting • Better security • Lower maintenance effort • Reduced storage costs • Better scalability • Real-time insights

Overall, Microsoft Fabric helps organizations manage data more efficiently and make better business decisions.

Thank You.

I am happy to answer any questions.